AI23331 Fundamentals of Machine Learning

TITLE OF PROJECT REPORT

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report "**Bus Crowd Detection System Using YOLOv8**" is the bonafide work of "SUGANYA (230701351), SWATHI SARAVANAN (230701356), VETRISELVI.M (230701379)" who carried out the project work under my supervision.

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ABSTRACT

Public transportation overcrowding is a significant safety and operational challenge. This project presents an automated Bus Crowd Detection System that uses the YOLOv8 large model (YOLOv8l) to perform real-time person detection from bus video footage. The system captures video frames, resizes them for uniform processing, and applies deep learning-based object detection to count the number of persons visible in each frame. Based on a configurable bus capacity (default: 40), the system computes a crowd percentage and classifies the occupancy into three levels: Low Crowd (below 50%), Medium Crowd (50–79%), and Overcrowded (80% and above). Visual overlays including bounding boxes, confidence scores, crowd status, and real-time timestamps are rendered on each frame. An alert is prominently displayed when the bus is detected as overcrowded. The solution leverages OpenCV for video capture and display, and the Ultralytics YOLO framework for inference, demonstrating an effective, scalable approach to intelligent crowd monitoring in urban transit systems. This solution offers a scalable, costeffective, and deployable approach for smart transportation authorities to improve passenger safety, optimize fleet management, and reduce the risks associated with overcrowded public transit. Future enhancements may include mobile app integration and predictive analytics to forecast crowd patterns and improve decisionmaking.

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INTRODUCTION

Urban public transportation systems, particularly buses, are prone to severe overcrowding during peak hours. Overcrowded buses present serious safety hazards including risk of accidents, passenger discomfort, and violations of transport regulations. Traditional approaches to monitoring passenger counts rely on manual checking or simple sensor-based systems that lack adaptability and accuracy in complex real-world conditions.

Computer vision and deep learning have emerged as powerful tools for intelligent crowd analysis. Object detection models trained on large datasets can reliably identify human figures in various lighting conditions, occlusions, and camera angles. YOLOv8 (You Only Look Once, version 8) is a state-of-the-art, real-time object detection framework known for its high speed and accuracy, making it ideal for processing live video streams.

This project implements a Bus Crowd Detection System that processes video footage from bus cameras using YOLOv8l (the large variant for higher accuracy) to count passengers in real time. The system classifies crowd density into three levels and triggers a visual alert when the bus is overcrowded, supporting transit operators in making informed decisions to improve passenger safety and service quality.

Manual passenger counting in public buses is often inaccurate, time-consuming, and impractical, especially in large-scale transit networks where thousands of passengers travel daily. Conductors or drivers may not be able to maintain precise counts during peak hours, leading to overcrowding, safety risks, and inefficient fleet management.

Overcrowded buses can result in discomfort, increased chances of accidents, and violations of safety regulations. Therefore, there is a strong need for an automated, reliable, and real-time system that can monitor passenger density without human intervention. A vision-based approach using advanced object detection techniques provides an effective solution by continuously analyzing video feeds and extracting meaningful insights about passenger count and occupancy levels.

The primary objective of this project is to design and develop a real-time passenger detection and monitoring system using the YOLOv8 (You Only Look Once version 8) model, a state-of-the-art deep learning algorithm known for its speed and accuracy in object detection tasks. The system aims to process live or recorded video footage from bus cameras and accurately detect individuals within each frame. Based on the detected number of passengers and a predefined or configurable bus capacity, the system classifies the crowd level into three categories: Low (below 50% capacity), Medium (50–80% capacity), and Overcrowded (above 80% capacity). This classification enables quick understanding of the occupancy status and helps in decision-making.

In addition to detection and classification, the system is designed to provide a comprehensive visual interface by overlaying important information directly onto the video stream. This includes bounding boxes around detected passengers, realtime person count, calculated crowd percentage, current crowd status, and timestamps for each frame. Such visual annotations enhance clarity and make the system user-friendly for operators and monitoring personnel. Furthermore, a key feature of the system is its ability to trigger a clear and prominent visual alert whenever the occupancy exceeds safe limits, ensuring immediate attention and action from authorities.

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LITERATURE REVIEW

Several research works have explored crowd detection and passenger counting using computer vision techniques. Early approaches relied on background subtraction and optical flow methods, which were computationally efficient but struggled under complex backgrounds and varying lighting conditions.

Redmon et al. (2016) introduced the YOLO (You Only Look Once) architecture, which redefined real-time object detection by treating it as a single regression problem. Successive versions (YOLOv3, YOLOv4, YOLOv5, YOLOv8) improved accuracy and efficiency significantly.

Jocher et al. (2023) released Ultralytics YOLOv8, which introduced a new backbone, anchor-free detection head, and improved training strategies, achieving superior performance on standard benchmarks.

Zhang et al. (2019) proposed CSRNet, a deep learning model for congested crowd counting using dilated convolutions, showing high accuracy in dense crowd scenarios. However, such density estimation methods are best suited for fixed-angle surveillance cameras rather than moving bus cameras.

Li et al. (2021) demonstrated the effectiveness of YOLO-based person detection for transit passenger counting, reporting high precision under varying environmental conditions. Their work highlighted the advantage of bounding box-based counting over density map approaches for structured environments like buses and trains.

Mehta et al. (2022) integrated YOLOv5 with edge computing hardware for real-time bus passenger monitoring, achieving processing speeds suitable for live deployment.

Their system validated the feasibility of YOLO-based crowd detection in transportation contexts.

Wang et al. (2020) proposed a crowd counting method using convolutional neural networks with attention mechanisms, improving accuracy in highly dense and occluded environments. Their approach demonstrated robustness in challenging real-world scenarios, though it required high computational resources.

Bochkovskiy et al. (2020) introduced YOLOv4, which significantly enhanced detection performance by incorporating features like CSPDarknet53 and advanced data augmentation techniques. This model achieved a balance between speed and accuracy, making it suitable for real-time applications.

Tan et al. (2020) presented EfficientDet, a scalable and efficient object detection model that uses compound scaling and a bi-directional feature pyramid network (BiFPN). It achieved high accuracy with fewer parameters, making it suitable for resource-constrained environments.

The present work builds on these foundations by employing YOLOv8l for highaccuracy detection, combining real-time video processing with dynamic crowd level classification and an alert mechanism, specifically tailored for bus environments

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PROPOSED SYSTEM

The proposed Bus Crowd Detection System is a video analytics pipeline that integrates deep learning-based object detection with a rule-based crowd classification engine. The system takes a bus video as input, processes each frame using YOLOv8l to detect persons, computes crowd density metrics, and outputs an annotated video stream with real-time status information.

3.1 Architecture Diagram

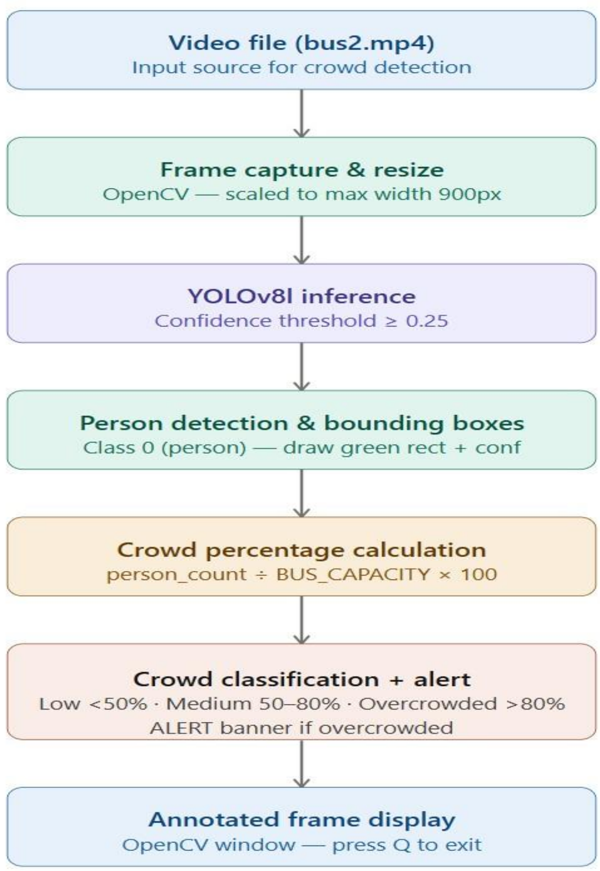

Figure 3.1 System Architecture Diagram

The proposed **Bus Crowd Detection System Using YOLOv8** is structured into multiple integrated modules, each responsible for a specific stage of video processing, detection, analysis, and real-time monitoring to ensure accurate and efficient crowd management in public buses. The system takes a bus video as input, processes each frame using YOLOv8l to detect persons, computes crowd density metrics, and outputs an annotated video stream with real-time status information.

1.Video Input Module

This module acts as the input source, where the system captures live camera feed or recorded video (e.g., bus2.mp4). It ensures continuous streaming of video data for further processing and analysis.

2. Frame Capture and Preprocessing Module

Implemented using OpenCV, this module extracts frames from the video and performs preprocessing steps such as resizing (scaled to a maximum width of 900px), normalization, and noise reduction to maintain consistent input quality for the detection model.

3.YOLOv8 Person Detection Module

This module utilizes the YOLOv8l deep learning model to detect persons in each video frame. It applies a confidence threshold (≥ 0.25) to filter detections and ensure only reliable predictions are considered, enabling accurate real-time object detection.

4. Bounding Box and Counting Module

Once persons are detected, this module draws bounding boxes around each individual and counts the total number of passengers present in the frame. Each detection is labeled with confidence scores for clarity and verification.

5. Crowd Percentage Calculation Module

This component calculates the crowd density by comparing the detected passenger count with the predefined bus capacity using the formula: Crowd Percentage = (Person Count / Bus Capacity) × 100.

This helps in quantifying occupancy levels in a precise and measurable way.

6. Crowd Classification Module

Based on the calculated percentage, this module classifies the crowd into three categories: Low (<50%), Medium (50–80%), and Overcrowded (>80%). This classification enables quick understanding of the bus occupancy status.

7. Annotated Output Display Module

The final module displays the processed video frames with all annotations, including bounding boxes, passenger count, crowd percentage, status labels, and timestamps. The output is shown in real time using an OpenCV window, allowing users to monitor the system effectively.

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MODULE DESCRIPTION

4.1 Module 1 – Video Capture and Frame Preprocessing

This module is responsible for reading video frames from the input file (bus2.mp4) using OpenCV's VideoCapture class. Before passing frames to the detection model, each frame is resized to a maximum width of 900 pixels while maintaining the original aspect ratio. This normalization ensures consistent processing speed regardless of the original video resolution. The module validates that the video file is opened successfully and terminates gracefully if the source is unavailable.

Key operations include: opening the video stream, reading frames in a loop, computing the scale factor for resizing, and applying cv2.resize(). A named window is created to display results in real time.

4.2 Module 2 – YOLOv8 Person Detection

This module performs the core detection task using the YOLOv8l pretrained model loaded via the Ultralytics YOLO library. For each resized frame, inference is executed with a confidence threshold of 0.25 to filter out low-confidence detections. The model outputs bounding boxes, class IDs, and confidence scores for all detected objects.

The module filters detections to retain only class ID 0, which corresponds to 'person' in the COCO dataset on which YOLOv8 is pretrained. For each detected person, a green bounding rectangle is drawn on the frame along with a text label showing the confidence score. The total person count is accumulated across all detections in the current frame.

4.3 Module 3 – Crowd Analysis and Alert Generation

This module computes the crowd percentage by dividing the detected person count by the configured bus capacity (BUS_CAPACITY = 40) and multiplying by 100. Based on the computed percentage, the system classifies the crowd into one of three levels:

- Low Crowd (Green): Crowd percentage below 50%.
- Medium Crowd (Yellow): Crowd percentage between 50% and 79%.
- Overcrowded (Red): Crowd percentage of 80% or above.

The module also captures the current system time and displays it along with key information such as passenger count, occupancy percentage, and crowd status directly on the video frame. In situations where the occupancy exceeds safe limits, the system highlights the condition with a clear and noticeable alert message, ensuring that operators can quickly recognize and respond to overcrowding. This module enhances situational awareness by combining real-time analytics with intuitive visual communication.

4.4 Module 4 – Annotated Output Display and User Interaction

This module is responsible for presenting the final processed video output to the user in real time. It displays each frame with all visual annotations, including bounding boxes, passenger count, crowd percentage, status labels, and timestamps using an OpenCV window. The module ensures smooth rendering of frames to maintain a continuous video stream experience. Additionally, it provides basic user interaction controls, allowing the user to terminate the application by pressing a designated key (such as 'Q').

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IMPLEMENTATION AND RESULTS

The experimental setup for the Bus Crowd Detection System Using YOLOv8 was designed to ensure accurate real-time person detection, efficient video processing, and reliable crowd classification under varying conditions. The setup consists of four key components — video dataset, deep learning model environment, processing pipeline, and output visualization system.

5.1 EXPERIMENTAL SETUP:

The dataset used for experimentation consists of bus surveillance video footage (bus2.mp4) containing varying passenger densities, lighting conditions, and movement patterns. The video frames are processed using the OpenCV library in Python, where each frame is extracted, resized, and normalized to maintain consistency in input resolution. Frame resizing is performed dynamically to a maximum width of 900 pixels while preserving the aspect ratio, ensuring optimal processing speed without significant loss of detail.

For the model environment, the YOLOv8l (You Only Look Once version 8 large) pretrained model provided by Ultralytics was utilized for person detection. The model was implemented in Python and executed in a local development environment with required dependencies such as OpenCV and the Ultralytics library. A confidence threshold of 0.25 was applied during inference to filter out lowconfidence detections and improve reliability. The model processes each frame individually and outputs bounding boxes, class labels, and confidence scores, from which only the ‘person’ class (class ID 0) is selected for counting.

The output visualization component is implemented using OpenCV display functions, where annotated frames are presented in a resizable window. The system overlays key information such as person count, crowd percentage, crowd status, and timestamp directly onto the video. Bounding boxes are drawn around detected individuals, and a prominent alert message is displayed when overcrowding conditions are detected. The system runs continuously until the video ends or the user manually terminates execution, ensuring real-time monitoring capability.

Overall, the experimental setup provides a robust and efficient framework for evaluating the system's performance in realistic scenarios. The integration of a highperformance detection model, optimized video processing pipeline, and clear visual feedback ensures accurate crowd monitoring and demonstrates the system's applicability in real-world public transportation environments.

5.2 RESULTS

The experimental evaluation of the Bus Crowd Detection System Using YOLOv8 demonstrated strong performance and reliability in real-time passenger monitoring and crowd assessment. The YOLOv8l model effectively detected and localized individuals in each video frame, even under varying lighting conditions and moderate occlusions. With a confidence threshold of 0.25, the system maintained a good balance between detection accuracy and processing speed, ensuring that most passengers were accurately identified without significant false positives. The bounding box-based counting approach proved highly effective for structured environments like buses, enabling precise estimation of passenger count. The crowd analysis module successfully converted this count into meaningful occupancy metrics by calculating the crowd percentage relative to the predefined bus capacity.

PROJECT SNAPSHOTS:

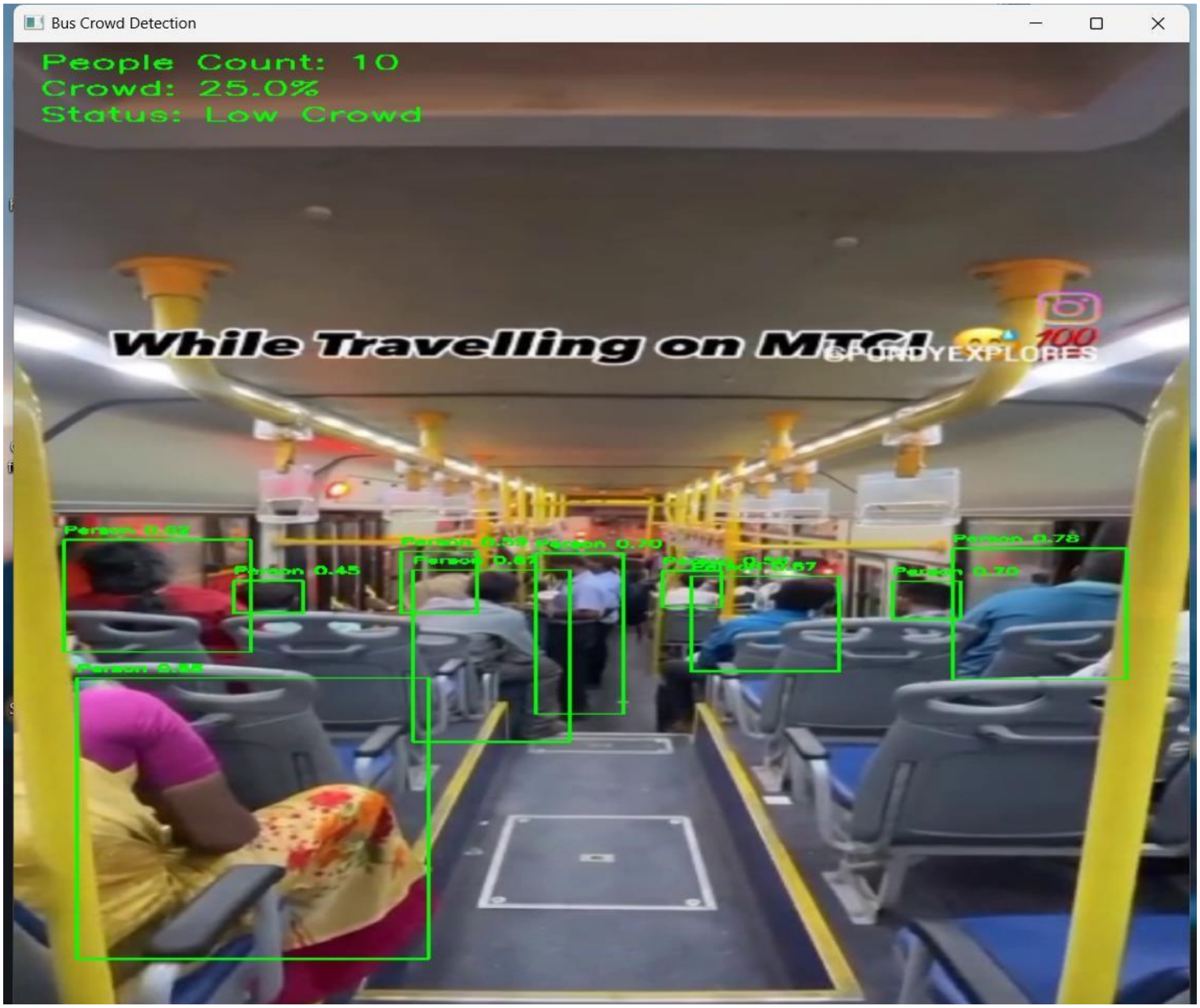

Figure 5.1 Real-Time Passenger Detection and Counting

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CONCLUSION AND FUTURE WORK

This project successfully developed a real-time Bus Crowd Detection System using the YOLOv8l deep learning model, capable of accurately identifying passengers from bus video footage and analyzing crowd density. The system processes each video frame to detect individuals, calculates occupancy as a percentage of the predefined bus capacity, and classifies the crowd level into Low, Medium, or Overcrowded categories. It also provides clear visual feedback through bounding boxes, annotated statistics, and alert messages when overcrowding conditions are detected. The integration of Python, OpenCV, and the Ultralytics YOLO framework resulted in an efficient and scalable pipeline suitable for continuous monitoring.

The system demonstrates the effectiveness of applying advanced computer vision techniques to real-world transportation challenges. By utilizing a pretrained YOLOv8 model, the solution achieves reliable detection performance without requiring extensive custom training data. The project highlights the potential of AIdriven surveillance systems in enhancing passenger safety, improving operational efficiency, and supporting smart public transport management. Overall, the system provides a practical and deployable solution for real-time crowd monitoring in buses.

The system processes each video frame to detect individuals, calculates occupancy as a percentage of the predefined bus capacity, and classifies the crowd level into Low, Medium, or Overcrowded categories. It also provides clear visual feedback through bounding boxes, annotated statistics, and alert messages when overcrowding conditions are detected.

The future scope of the Bus Crowd Detection System focuses on enhancing its realworld applicability, scalability, and performance. The system can be extended by integrating it with live CCTV or onboard camera feeds to enable continuous realtime monitoring in public transportation. Implementing advanced passenger tracking mechanisms can further improve counting accuracy, especially during entry and exit movements.

Optimization of the model using lightweight YOLO variants such as YOLOv8n or YOLOv8s can support deployment on embedded edge devices within buses. Incorporating data logging and analytics features will help in analyzing historical crowd patterns and assist in better decision-making for transit planning. Furthermore, improving detection performance in low-light or night conditions using image enhancement techniques or infrared cameras can make the system more robust and reliable in diverse operating environments.

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